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Dual Semantic Guidance for Open Vocabulary Semantic Segmentation

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Abstract

Open-vocabulary semantic segmentation aims to enable models to segment arbitrary categories. Currently, though pre-trained Vision-Language Models (VLMs) like CLIP have established a robust foundation for this task by learning to match text and image representations from large-scale data, their lack of pixel-level recognition necessitates further fine-tuning. Most existing methods leverage text as a guide to achieve pixel-level recognition. However, the inherent biases in text semantic descriptions and the lack of pixel-level supervisory information make it challenging to fine-tune CLIP-based models effectively. This paper considers leveraging image-text data to simultaneously capture the semantic information contained in both image and text, thereby constructing Dual Semantic Guidance *and corresponding pixel-level pseudo annotations. Particularly, the visual semantic guidance is enhanced via explicitly exploring foreground regions and minimizing the influence of background. The dual semantic guidance is then jointly utilized to fine-tune CLIP-based segmentation models, achieving decent fine-grained recognition capabilities. As the comprehensive evaluation shows, our method outperforms state-of-art results with large margins, on eight commonly used datasets with/without background.*

1. Introduction

Open-Vocabulary Semantic Segmentation (OVSS) aims to segment arbitrary categories in open environments, distinguishing itself from traditional semantic segmentation [1–3], which operates within a predefined set of categories. Recently, Visual-Language Models (VLMs) such as CLIP [4] and Align [5] demonstrate exceptional capabilities in aligning textual descriptions with corresponding images, making them highly suitable for open-vocabulary tasks. However, these VLMs with image-level visual-text alignment lack fine-grained expressive power, resulting in poor segmentation precision when directly applied to the OVSS task. Currently, several approaches [6–10] train segmentation models based on VLMs using image-text data which is rich in semantic information. This enables the models to acquire fine-grained segmentation capabilities while retaining their generalization ability across open-vocabulary categories, which provides a promising strategy for tackling semantic segmentation tasks in open-vocabulary scenarios.

Let's look deeper in open-vocabulary semantic segmentation tasks utilizing VLMs: the core principle involves aligning pixel-level features with text embeddings, and assigning each pixel a category based on these alignments. Currently, some methods [6, 8] trained on image-text data rely on the VLM model's ability to align part of images with the complete texts, establishing weak constraints for segmentation training. In contrast to employing complete texts, some methods [7, 9] attempt to extract nouns from texts to obtain category semantic information that matches local regions in images. However, nouns lacking detailed descriptions can suffer from "Inherent biases", which arise from the semantic disparity between coarse-grained text and corresponding images. In image-text data, the nouns in text often offers coarse-grained information, where "cat" and "dog" are both labeled as "animal" (Fig 1.a), yet there are substantial differences in image level. Moreover, some existing methods [6, 9] lack the pixel-level semantic supervisory information to fine-tune VLMs. As a result, poorly segmented images can still align with the corresponding text well, as shown in Fig 1.b.

In this paper, we denote the semantic information used for matching images pixels as **semantic guidance**. We identify that the crucial factor in improving the model's fine-grained segmentation performance is the establishment of granular semantic guidance (at both text and image level) and accurate segmentation annotations. To acquire more precise semantic guidance and segmentation labels, we explore the feasibility of extracting semantic guidance from images which are abundant in semantic content. The most direct form is the various instances within an image, as these instances provide an unbiased semantic representation of the corresponding regions. By utilizing foundation models such as SAM [11], we can efficiently extract a large amount

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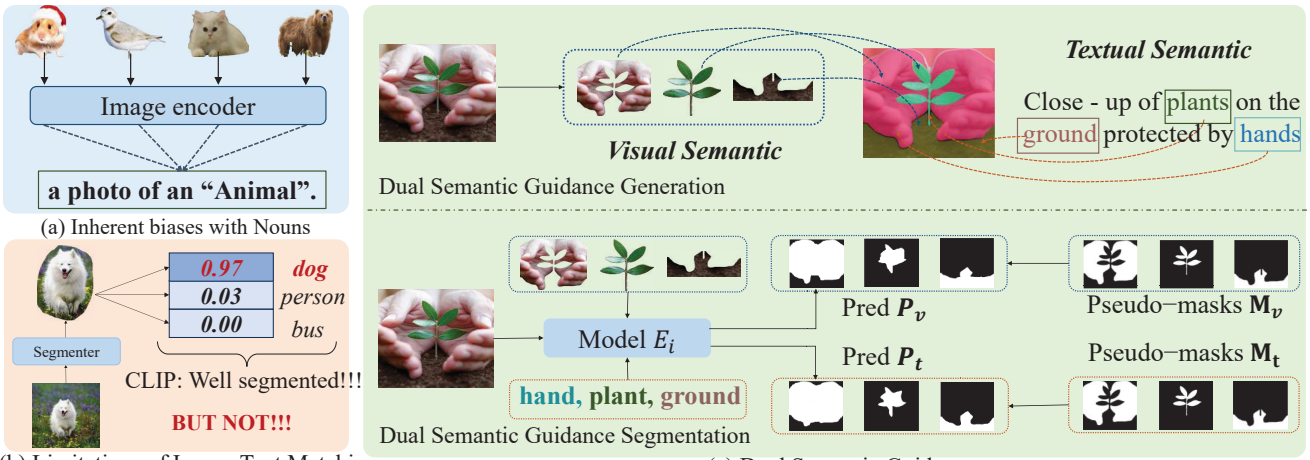

(a) Inherent biases with Nouns

(b) Limitations of Image-Text Matching

(c) Dual Semantic Guidance

Figure 1. Previous methods shortcomings and overview of Dual Semantic Guidance. (a) Relying solely on nouns leads to the convergence of image representations for subcategories, resulting in inherent bias. (b) The rough segmentation, or even the unsegmented dog, is semantically closest to the "dog" category, highlighting the limitations of using image-text matching to supervise segmentation. (c) Our method captures dual semantic guidance from image-text pairs, collaboratively guiding the model's training.

of visual semantic guidance from each image, and without the need for manual annotation. However, relying solely on visual semantic guidance deviates from the pre-training of VLMs and suffers from modality domain gap. To ensure inter-modal consistency within the VLM model, we also extract nouns from text and match them with the instances to form textual semantic guidance. The dual-modality semantic guidance mechanisms harness the comprehensive information within image-text data, synergistically improving fine-grained recognition capability of the VLM model.

Specifically, we propose *Dual Semantic Guidance* to extract semantic information from both visual and textual perspectives within image-text data. The pipeline consists of *Dual Semantic Guidance Generation* (DSGG) and *Dual Semantic Guidance Segmentation* (DSGS). In DSGG, we utilize SAM to extract instances from images, and enhance visual semantics to focus more on instance through a training-free Region Awareness Attention module. Subsequently, through clustering and filtering, we attain high-quality, independent visual semantic guidance. On the other hand, independent nouns are extracted from the text, which are then matched with the visual semantic guidance to construct textual semantic guidance. In DSGS, we design a training strategy for dual semantic guidance to collaboratively enhance the model's fine-grained representation capabilities. Additionally, we develop the Background Screening module, which integrates the image feature map with the semantic guidance to jointly predict the image background. Our method surpasses state-of-the-art results on various test datasets, including coco-stuff [12], ADE [13], etc., achieving an average mIoU of 48.3 on three datasets with background, and 42.8 on five datasets without background.

2. Related Work

2.1. Vision Foundation Models

The latest advancements in foundational vision models demonstrate that large-scale pre-training techniques have facilitated the development of potent and versatile models [5, 11]. These models exhibit remarkable adaptability across various tasks [14–19]. The emergence of highly expressive vision-language models, such as Align and CLIP, has been witnessed in the realm of large-scale contrastive pre-training of text-image pairs. Furthermore, SAM [11] has undergone iterative training on billions of weakly annotated images, achieving remarkable segmentation performance, but it lacks semantic discriminative capabilities. Researchers have proposed various integration strategies: SAM-CLIP [20] constructs a unified model that skillfully integrates the functionalities of SAM and CLIP into a single framework. Open-Vocabulary SAM [21] integrates a CLIP encoder with a SAM decoder, bridging these two distinct components through knowledge distillation. Additionally, some methods [22–24] integrate multiple models [25–27] to collaboratively form a powerful open-vocabulary segmenter. However, such methods often demand substantial computational resources.

Compared to these workflow-based integrated models, our approach relies solely on CLIP's image encoder as the segmenter during the testing phase, achieving open-vocabulary segmentation without adding any additional modules.

2.2. Open-Vocabulary Semantic Segmentation

The emergence of Vision-Language Models (VLMs) with aligned image-language representations has brought new perspectives to open-vocabulary tasks. However, these VLM models prioritize the alignment between the holistic image and its textual description, overlooking the fine-grained discriminative capabilities of individual pixels. Some works [28–30] employ supplementary modules and mechanisms to enhance the VLM's segmentation capabilities, which entail substantial annotation costs. On the other hand, some studies have focused on developing open-vocabulary segmentation capabilities from weakly labeled or even unlabeled data, categorized into text-supervised semantic segmentation and additional model-guided semantic

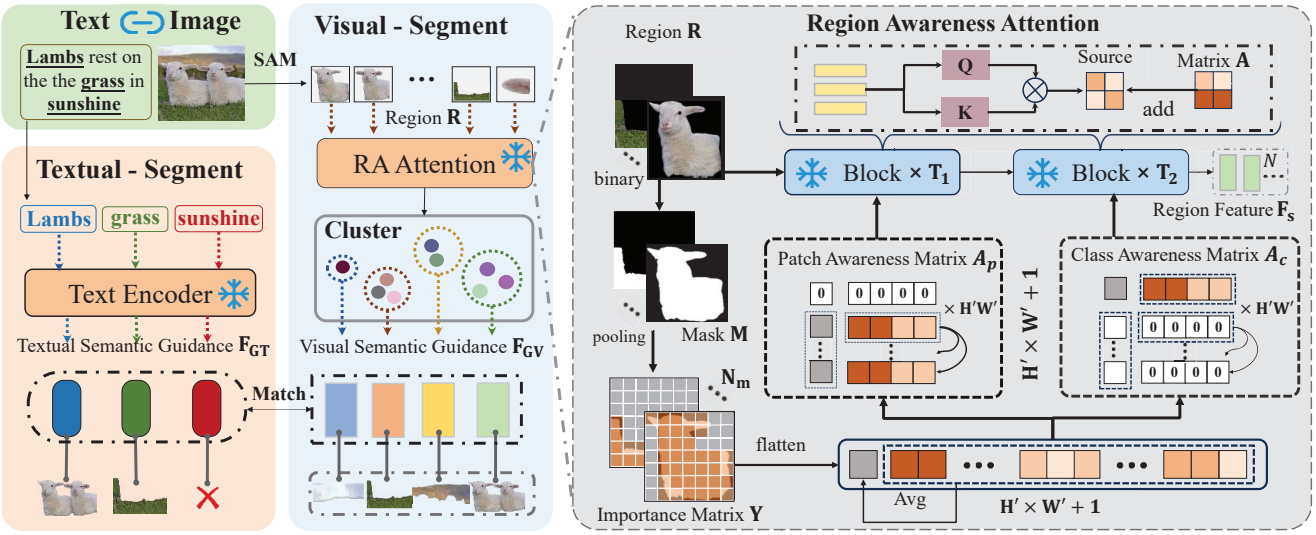

Figure 2. Dual Semantic Guidance Generation (DSGG). We utilize SAM to obtain coarse masks, followed by extracting foreground-enhanced features via "Region Awareness Attention". The features are clustered and filtered to yield visual semantic guidance. We extract nouns from the text to identify textual semantic guidance. H' and W' are the input resolutions of the image encoder.

segmentation based on the differences in training data.

The text-supervised methods primarily focus on using text to establish semantic segmentation objectives, leveraging image-text matching to supervise model training: GroupViT [7] matches text with clustered image regions. SegCLIP [31] augments CLIP with an inserted grouping module, relying on text to introduce additional supervision; PGSeg [10] introduces non-learnable prototype regularization to optimize GroupViT; TCL [6] utilizes region-text matching as a supervisory signal; CoDe [9] decomposes text into noun phrases, focusing on matching image regions with nouns. The model-guided methods employ additional models to provide segmentation pseudo-labels, CLIP-DINOiser [32] utilizes the DINO [27] to generate coarse masks, enforcing semantic consistency within the masks; PixelCLIP relies on SAM [11] to provide regional segmentation results and employs learnable text features as guidance to supervise model training.

Most of the aforementioned methods rely on text while overlooking the potential of image instances as high-quality visual semantic guidance to align model outputs. In case of lacking pixel-level annotations, their performances are sub-optimal. Our approach focuses on constructing dual semantic guidance from both image and text, generating pixel-level pseudo-labels to jointly facilitate the adaptation of the CLIP model to dense visual prediction tasks.

3. Dual Semantic Guidance

3.1. Method Overview

Our method consists of two components: Dual Semantic Guidance Generation (DSGG), which produces visual and textual semantic guidance as dual semantic guidance for model training, and dual Semantic Guidance Segmentation (DSGS), which leverages the Dual semantic guidance to jointly enhance the model's segmentation performance, as illustrated in Figures 2 and 4, respectively.

Dual Semantic Guidance Generation We process image-text data in two parallel streams. On one hand, we input the image $\mathbf{I}$ from image-text data into SAM to produce fragmented segment regions. Then we process them through Region Awareness Attention and Clustering to generate $\mathbf{K}_v$ independent visual semantic guidance $\mathbf{F}_{GV} \in \mathbb{R}^{\mathbf{K}_v \times C}$ along with their corresponding pseudo-masks $\mathbf{M}_v \in \mathbb{R}^{H \times W \times K_v}$. On the other hand, we extract nouns from the text, process them via a text encoder, and subsequently match and filter them against the visual semantic guidance, ultimately generating $\mathbf{K}_t$ textual semantic guidance $\mathbf{F}_{GT} \in \mathbb{R}^{\mathbf{K}_t \times C}$ with their corresponding pseudo-masks $\mathbf{M}_t \in \mathbb{R}^{H \times W \times \mathbf{K}_t}$.

Dual Semantic Guidance Segmentation During the training phase, we obtain the feature map $\mathbf{F}_i$ of each image via image encoder and compute the similarity with textual or visual semantic guidance to respectively derive the prediction $\mathbf{P}$, while calculating the loss based on pseudo-mask. Concurrently, Additionally, we provide supervision for training the background screening module. During the inference stage, we rely solely on the image encoder E_i, and text encoder E_t to extract image feature map $\mathbf{F}_i$ and category features. By calculating the similarity, we obtain the final segmentation results $\mathbf{P}$. Notably, the use of Background Screening remains optional.

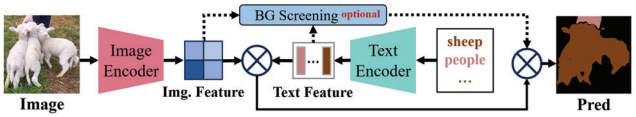

Figure 3. Inference flow

3.2. Dual Semantic Guidance Generation

We process image and text separately. For image, we employ SAM to automatically generate numerous fragmented

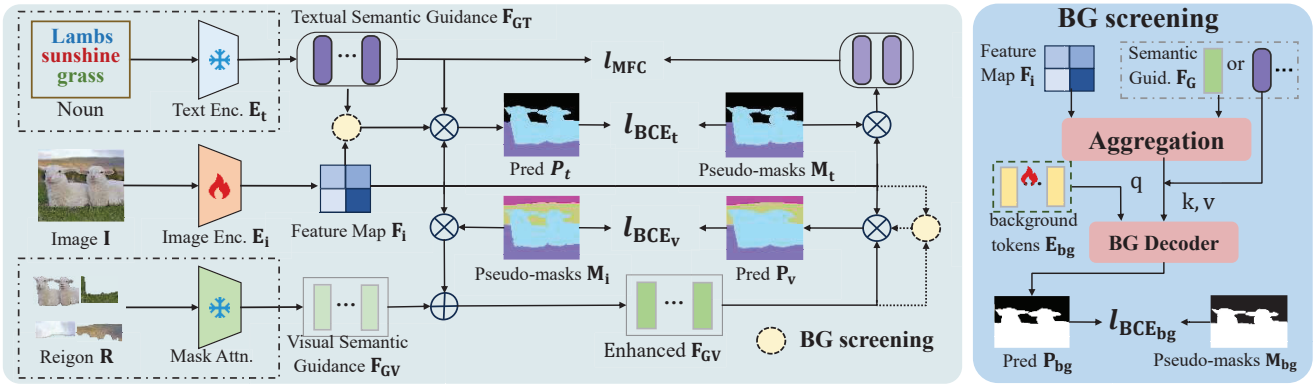

Figure 4. Dual Semantic Guidance Segmentation (DSGS). (a) DSGS encodes image to feature map F_i, and calculates its similarity to the dual semantic guidance $\mathbf{F_{GV}}/\mathbf{F_{GT}}$ to ascertain pixel classifications. Distinct training strategies are employed for different semantic guidance. (b) The Background screening jointly assesses the background of the image from both spatial and categorical perspectives using the semantic guidance $\mathbf{F_{GV}}/\mathbf{F_{GT}}$ and feature map F_i.

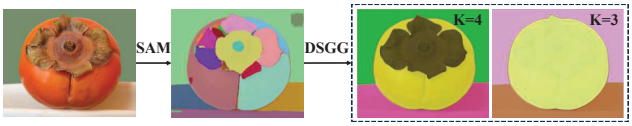

Figure 5. Demonstrates the aggregation results of the DSGG on disorganized SAM segmentation results. Notice that, when K=4, it includes all instances from K=3 plus the "leaf" instance.

Figure 6. The irregularity and unnatural nature of the background cause CLIP to erroneously encode the foreground information, whereas Region Awareness Attention can rectify it.

regions for images $\mathbf{I}$. We initially filter these regions $\mathbf{R}$ based on the confidence scores output by SAM, and then retain the top $\mathbf{N}_m$ regions with the largest areas. Using the Region Awareness Attention introduced later, we obtain the region feature set $\mathbf{F}_s \in \mathbb{R}^{\mathbf{N}_m \times C}$ for all regions. Subsequently, we apply K-means clustering to filter and merge $\mathbf{F}_s$ to generate $\mathbf{K}$ clusters. We select $\mathbf{K}_v$ clusters as the visual semantic guidance, denoted as $\mathbf{F}_{GV} \in \mathbb{R}^{K_v \times C}$. The Pseudo-masks for $\mathbf{F}_{GV}$ denote as $\mathbf{M}_v$. Fig. 5 shows the results.

For text, we use NLTK [33] to identify and extract nouns from text, and encode them through a CLIP text encoder E_t to obtain features $\mathbf{F}_{Tm}$. To ensure high semantic matching, we employ Eq.(1) to match and filter $\mathbf{F}_{Tm}$ with $\mathbf{F}_{GV}$. Additionally, to guarantee semantic uniqueness, we utilize Eq.(2) for joint filtering, thereby obtaining textual semantic guidance $\mathbf{F}_{GT} \in \mathbb{R}^{K_t \times C}$, with its Pseudo-masks $\mathbf{M}_t$.

$$\left(Softmax\left(F_{GV}(:,:) \cdot F_{Tm}^{T}(j,:)\right)\right)_i > T_A, \tag{1}$$

$$\left(F_{GV}(:,:) \cdot F_{Tm}^{T}(j,:)\right)_i > T_B, \tag{2}$$

where thresholds $T_A = 0.8$ and $T_B = 36$.

Region Awareness Attention For each instance segmented by SAM, the instance region is "foreground" and other areas are "background". CLIP tends to misclassify polygon instances due to background interference (mistaking "sky" for "airplane" in Fig.6). We introduce RAA to focus feature extraction on the foreground. Firstly, we enhance the encoder's focus on the foreground areas within the regions, minimizing the influence of background and instance edges. Here, we define the importance of each patch as proportional to the number of foreground pixels it contains. Given an image region and its corresponding mask $\mathbf{M}$, we utilize average pooling to obtain the importance matrix $\mathbf{Y}$ from its binary mask as:

$$\mathbf{Y}_{i,j} = \frac{1}{\mathbf{S}^2} \sum_{m=0}^{\mathbf{S}-1} \sum_{n=0}^{\mathbf{S}-1} \mathrm{M}[\mathbf{S} \times i + m,\ \mathbf{S} \times j + n], \tag{3}$$

where $\mathbf{S} = \mathbf{16}$ denotes the stride. The importance matrix $\mathbf{Y}$ is then normalized to the range of [0, 2]. We consider that during the early stage of encoding, the model should focus on self-awareness among patches to thoroughly explore the foreground regions. Furthermore, to guide the model's judgment of regions $\mathbf{R}$ towards the foreground areas in the later stage, we want to enhance the class token's attention to these regions. For these purposes, we construct two distinct matrix, the Patch Awareness Matrix $\mathbf{A}_p$ and the Class Awareness Matrix $\mathbf{A}_c$ as follows:

$$\begin{aligned} \mathbf{A}_p &= concat\{[0,\ldots,0],\ [Avg(\mathbf{Y}),\ \mathbf{Y}]_{repeat(H' \times W')}\}, \\ \mathbf{A}_c &= concat\{[Avg(\mathbf{Y}),\ \mathbf{Y}],\ [0,\ldots,0]_{repeat(H' \times W')}\}, \end{aligned} \tag{4}$$

where H' and W' are the input resolutions of the image encoder. The constructed awareness matrix $\mathbf{A}(\mathbf{A}_p,\ \mathbf{A}_c)$ are integrated into the attention mechanism as:

$$Attention(Q, K, V) = Softmax\left(\frac{QK^{\top}}{\sqrt{d}} + A\right)V, \tag{5}$$

Ultimately, we obtain the features $\mathbf{F}_s$ of all regions $\mathbf{R}$. Figure 6 shows the efforts of the Region Awareness Attention.

3.3. Dual Semantic Guidance Segmentation

We utilize ViT-based MaskCLIP as the image encoder to extract an image feature map $\mathbf{F}_i$ from the image. Additionally, we design the Background Screening module, which combines semantic information and the feature map to predict the background region $\mathbf{P}_{bg}$. $\mathbf{P}_{bg}$ is subsequently fused with the overall prediction results $\mathbf{P}_r \in \mathbb{R}^{H\times W\times(\mathrm{N}+1)}$, where $\mathbf{N}$ is the number of semantic guidance. For textual semantic guidance, the prediction are derived by calculating the cosine distance with image features map $\mathbf{F}_i$, and then compute the Binary Cross-Entropy (BCE) loss with corresponding pseudo masks. However, coarse-grained text features impair the model's expression of fine-grained semantic information in images, which leads to Inherent biases. It is not the optimal direction for pixel features. Therefore, we introduce the Masked Feature Contrastive loss (MFC loss) to ensure the average semantic representation within the mask diverges from negative samples, avoiding optimizing towards coarse-grained text features solely. We define MFC loss as:

$$\begin{aligned}\mathcal{L}_{\mathrm{MFC}}(\mathbf{S}) = &-\frac{1}{2B}\sum_{k=1}^{B}\log\frac{\exp(S_{k,k}/\tau)}{\sum_{j=1}^{B}\exp(S_{k,j}/\tau)}\\ &-\frac{1}{2B}\sum_{k=1}^{B}\log\frac{\exp(S_{k,k}/\tau)}{\sum_{j=1}^{B}\exp(S_{j,k}/\tau)},\end{aligned} \tag{6}$$

$$S_{k,j} = \left(\frac{F_i \times M}{\sum_{h=0}^{H}\sum_{w=0}^{W} M(h,w)}\right)^{\mathrm{T}} \cdot F_{GT}, \tag{7}$$

where τ is a learnable temperature, $\mathbf{M}$ denotes the 0-1 pseudo masks, $S_{k,j}$ stores the cosine similarity between the average representation of k-th instance region and the j-th textual semantic guidance.

For the visual semantic guidance, it represent fine-grained semantic expressions and serve as the optimal direction for each pixel feature. In Eq. 8, we perform a weighted average based on the similarity between the feature map $\mathbf{F}_i$ and $\mathbf{F}_{GV}$ to obtain the instance semantic representation within the mask Mv. It is then fused with the $\mathbf{F}_{GV}$ by averaging, guiding pixel feature optimizing within the mask. This operation is termed as Guidance Enhance. We employ BCE loss to penalize the prediction with corresponding pseudo masks.

$$F_{GV} = \frac{1}{2}\left(\sum\frac{(F_i^T \cdot F_{GV})\times M}{\sum(F_i^T \cdot F_{GV})\times M}\cdot F_i + F_{GV}\right). \tag{8}$$

Background Screening To better align with the open-vocabulary definition, we designate all regions not belonging to the semantic guidance (target we not aim to segment) as background. The background screening module is responsible for background reasoning, fusing semantic guidance F_G and contextual information of the image feature map $\mathbf{F}_i$ to jointly predict the background $\mathbf{P}_{bg}$, rather than relying on saliency detection [34]. During Aggregation [35–37], we calculate the cosine similarity between $\mathbf{F}_i$ and $\mathbf{F}_G$ to obtain initial cost map $\mathbf{F}_{cv}$, as shown in:

$$\mathbf{F}_{cv}(i,j,n) = \frac{\mathbf{F}_i(i,j)\cdot\mathbf{F}_G(n)}{\|\mathbf{F}_i(i,j)\|\cdot\|\mathbf{F}_G(n)\|}, \tag{9}$$

where i, j indicate the 2D spatial position, n represents the index of semantic guidance. Then the initial cost map is passed through a convolution layer to generate an updated cost map $\mathbf{F}_{cv} \in \mathbb{R}^{H'\times W'\times \mathrm{N}\times D}$, where $\mathbf{N}$ is the number of semantic guidance. To endow the cost map with spatial awareness, we input the cost map $\mathbf{F}_{cv}$, fused with guidance from $\mathbf{F}_i$ [38]. Attention is computed across pixels to understand the spatial correlations in the image data, and $\mathbf{F}_{cv}$ is updated as the input for the next step:

$$\mathbf{F}_{cv}(:,:,n,:) = \mathcal{T}(\mathbf{F}_{cv}(:,:,n,:)), \tag{10}$$

where n represents the index of semantic guidance, $\mathcal{T}$ represents the transformer block.

We utilize a decoder to explore the relationships among diverse semantic representations of pixels and subsequently predict the background. We integrate F_{cv}, guided by semantic guidance [38], with learnable background tokens shared among pixels, through a transformer decoder to obtain the background prediction $\mathbf{P}_{bg} \in \mathbb{R}^{H'\times W'}$:

$$\mathbf{P}_{\mathrm{bg}}(i,j) = \mathcal{D}\left(\mathbf{T}_{bg}(i,j,:,:),\ \mathbf{F}_{cv}(i,j,:,:)\right), \tag{11}$$

where $\mathbf{T}_{bg}$ represents the learnable background tokens, $\mathcal{D}$ represents a decoder.

3.4. Supervision for Network Training

The primary loss in our method comprises the binary cross-entropy (BCE) loss for two modality semantic guidance, the contrastive loss for textual semantic guidance, and the BCE loss for the Background screening. We formulate the overall training objective $\mathcal{L}$ as:

$$\mathcal{L} = \lambda_I\mathcal{L}_{\mathrm{BCE}_i} + \lambda_T\mathcal{L}_{\mathrm{BCE}_t} + \mathcal{L}_{\mathrm{MFC}} + \mathcal{L}_{\mathrm{BCE}_{\mathrm{bg}}}, \tag{12}$$

where $\lambda_I = 2.5$, $\lambda_T = 2.0$ empirically.

4. Experiments

4.1. Experimental Datasets

Training datasets: We use CC3M as training dataset. CC3M comprises 3 million image-text pairs and has been extensively used for training text-supervised semantic segmentation methods. We randomly sample $10\%(300,000)$ of the image-text pairs for training.

Evaluation datasets: We evaluate our model on five commonly used semantic segmentation datasets without background, including COCO Stuff [12], ADE20K [13], PASCAL Context59 [44], PASCAL VOC20 [45], and CityScapes [46], as well as three semantic segmentation datasets with background, including PASCAL Context60 [44], PASCAL VOC21 [45], COCO Object [12]. The

Table 1. We tested the performance of our method using mIoU on eight semantic segmentation benchmarks. All the results have no post-processing. Our method achieves state-of-the-art performance across various datasets.

Method	Publication	w/ background category			Avg.	w/o background category					Avg.
		VOC21	Ctx60	Object		VOC20	Ctx59	Stuff	ADE	City	
CLIP	ICML 2021	18.8	9.9	8.1	12.3	49.4	11.1	6.7	3.1	6.5	15.4
ViL-Seg [8]	ECCV 2022	37.3	18.9	18.1	24.8	-	-	-	-	-	-
ViewCo [39]	ICLR 2023	52.4	23.0	23.5	33.0	-	-	-	-	-	-
SimSeg [40]	CVPR 2023	57.4	26.2	29.7	37.8	-	-	-	-	-	-
OVSegmentor [41]	CVPR 2023	53.8	20.4	25.1	33.1	-	-	-	5.6	-	-
TCL [6]	CVPR 2023	51.2	24.3	30.4	35.3	77.5	30.3	19.6	14.9	23.5	33.2
SegCLIP [31]	ICML 2023	52.6	24.7	26.5	34.6	-	-	-	-	-	-
CoCu [42]	NeurIPS 2023	40.9	21.2	20.3	27.5	-	-	13.6	11.1	15.0	-
PGSeg [10]	NeurIPS 2023	53.2	23.8	28.7	35.2	-	-	-	-	-	-
CoDe [9]	CVPR 2024	46.0	23.1	25.4	31.5	64.9	28.2	17.6	12.9	24.4	29.6
CLIP-DINOiser [32]	ECCV 2024	62.1	32.4	34.8	43.1	80.9	35.9	24.6	20.0	31.7	38.6
PixelCLIP(ViT) [43]	NeurIPS 2024	-	-	-		**85.9**	37.9	23.6	18.7	27.2	38.7
PixelCLIP(CNN) [43]	NeurIPS 2024	-	-	-		67.2	35.4	21.4	20.3	34.8	35.8
Ours	CVPR 2025	**67.7**	**40.8**	**36.3**	**48.3**	85.0	**46.5**	**25.7**	**20.7**	**36.0**	**42.8**

mean Intersection over Union (mIoU) is employed as the evaluation metric.

4.2. Implementation Detail

We generate a vast number of masks via SAM ViT-B, and utilize the part-of-speech tagging algorithm from NLTK [33] for noun selection. For K-Means [47] clustering, we set k = 5. For the segmentation model, we employ MaskCLIP [48] with ViT-B/16. For Region Awareness Attention, we utilize CLIP with ViT-B/16. During the experiments, we set total batchsize = 128 with a initial learning rate = 6.4e-4, and 30,000 iterations are conducted (we open the last 6 layers of image encoder for training). Furthermore, we ultimately train the model for additional 1,000 iterations on textual semantic guidance alone to adapt it to the textual modality received during inference. The cosine schedule [49] is adopted to adjust the learning rate, and the weight decay of the AdamW optimizer [50] is set to 0.05. The model is trained on two RTX 3090 GPUs for five hours.

In addition, we solely use image encoder for the datasets without background (w/o bg); whereas for datasets with background (w/ bg), we additionally activate the Background Screening for background prediction.

4.3. Comparison with SOTA methods

In Table 1, we summarize and compare our method with existing zero-shot semantic segmentation models [6, 8–10, 31, 32, 39–43]. For a fair comparison, we do not employ any post-processing strategies to refine or smooth segmentation masks, such as PAMR [51] and DenseCRF [52]. Consequently, we re-evaluate CoDe [9] without PAMR. Our method outperforms the SOTA on three datasets with background (VOC21 +5.6%, Ctx60 +8.4%, Object +1.5%), achieving an average improvement of +5.2% in mIoU performance. Furthermore, our method outperforms the SOTA on four datasets without background (Ctx59 +8.6%, Stuff +1.1%, ADE +0.4%, Cityscape +1.2%), with an average performance improvement of +6.1% mIoU. Also note that our performance on VOC20 is second only to the SOTA.

Table 2. Analysis of each stage of Semantic Guidance Generation

Method	w/ BG	w/o BG
Ours	**48.3**	**42.8**
w/o Initial Filter	11.8	15.4
w/o Region Awareness Attn.	45.6	39.9
w/o Cluster	46.7	40.8

Figure 7 and 8 show the segmentation results from different methods. Since PixelCLIP [43] do not report its performance on datasets with background, we only compared it on the dataset with background. It is obvious that our method achieves better visual performance in both pixel classification and segmentation boundary, validating the effectiveness of Dual Semantic Guidance in segmentation tasks.

4.4. Ablation Study

Analysis of Semantic Guidance Generation In Table 2, we assess the efficacy of DSGG by conducting an ablation study on the three core steps: initial filter, Region Awareness Attention, and cluster. The initial filter stage effectively eliminates a considerable portion of low-quality instances that are often fragmented and devoid of meaningful semantic information, featuring only abrupt color edges. Omitting this stage (see "w/o Initially Filter") would expose the model to a significant amount of irrelevant noise during training. Region Awareness Attention provides accurate feature representations for each instance. The absence of this module (see "w/o Region Awareness Attn.") hinders the model's capacity to effectively capture instance features, subsequently impacting clustering and the semantic expression of visual semantic guidance, ultimately deteriorating model performance. The clustering consolidates semantically similar instances, shifting the segmentation emphasis to semantic categories and purging noisy data, ultimately yielding superior training data.

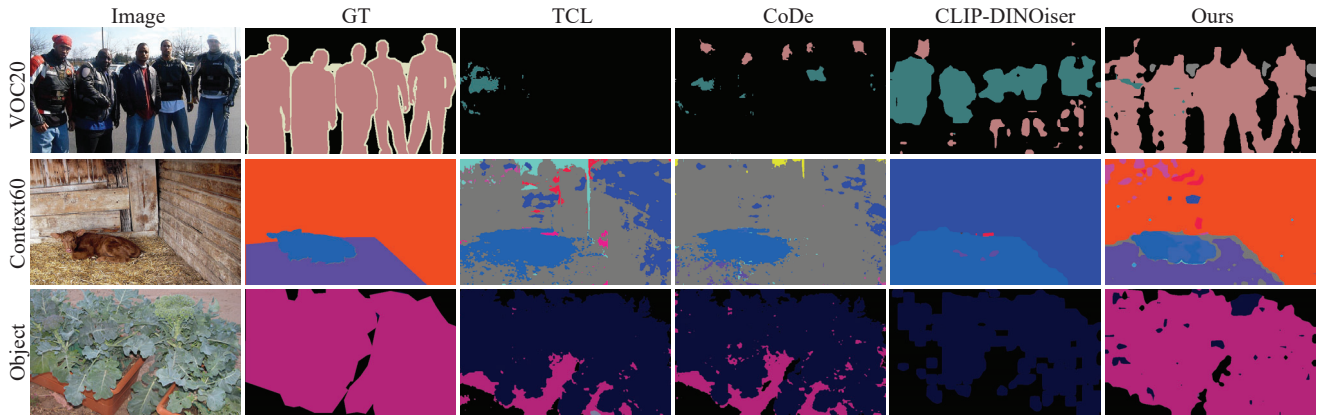

Figure 7. Segmentation results with background. We compare ours against TCL [6], CoDe [9], CLIP-DINOiser [32].

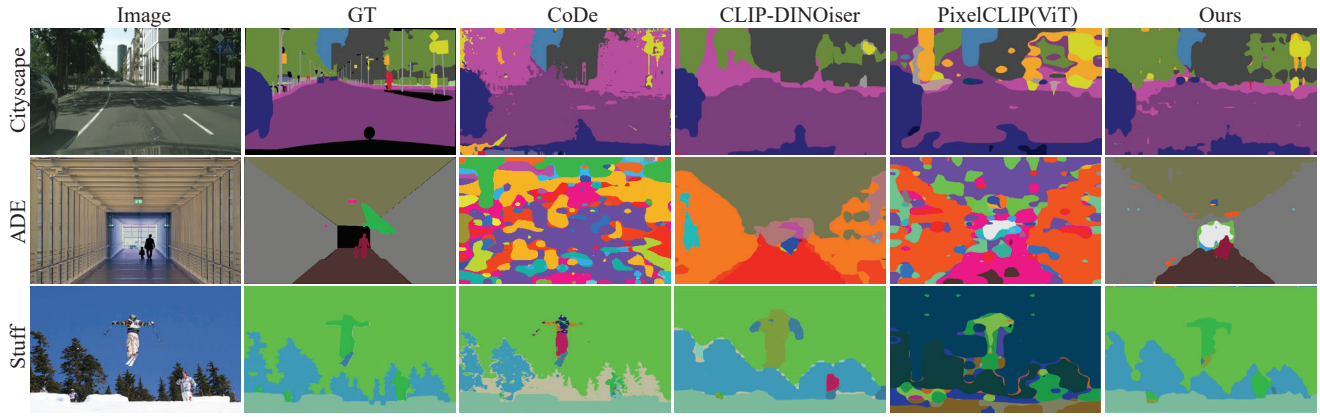

Figure 8. Segmentation results w/o background. We compare ours against CoDe [9], CLIP-DINOiser [32], PixelCLIP [53].

Table 3. Analysis of Components in Region-Specific Attention.

Method	w/ BG	w/o BG
Ori. CLIP	45.6	39.9
+ Patch attn.	46.5	40.6
+ Class attn.	47.3	41.8
+ C/P (All Stage)	47.1	42.3
+ C/P (Diff. Stage)	**48.3**	**42.8**

Analysis of Region Awareness Attention In Table 3, we evaluate the individual contributions of the components within RAA. Firstly, we use the original CLIP to encode the image segment(see "Ori. CLIP"), but because of the interference of the background area on the instance feature expression, it leads to lower performance. Secondly, we ablate the Patch Attention and Class Attention attention separately to validate effectiveness (see "+ Patch attn." and "+ Class attn."). The Patch Attention bolstered the self-awareness of regional tokens towards foreground areas, facilitating a more thorough capture of instance details. Meanwhile, the Class Attention steered the model's attention towards foreground regions, thereby enhancing the encoding of instance features. Finally, we test the simultaneous activation of Class Attention and Patch Attention across all blocks (+ C/P (All Stage)), resulting in lower performance compared to staged method, as the latter can focus on foreground regions while retaining global information.

Table 4. Analysis of Dual Semantic Guidance Segmentation.

Guid.	Method	w/ BG	w/o BG
Textual	Baseline	36.8	33.6
	+ MFC Loss	38.3	34.7
	+ Guid. Enhance	12.3	33.9
Visual	Baseline	28.8	39.9
	+ MFC Loss	27.7	38.5
	+ Guid. Enhance	30.8	42.5
Dual	Ours(Thresh 0.4)	46.5	**42.8**
	Ours(Bg screening)	**48.3**	**42.8**

Analysis of Dual Semantic Guidance Segmentation In table 4, we conduct ablation studies on both the visual and textual semantic guidance, as well as two components: the Guidance Enhance and Masked Feature Contrastive Loss(see in sec.3.3). Firstly, we observe that using only textual semantic guidance for training(see "Textual".) outperforms visual semantic guidance on datasets containing background classes. This advantage arises because textual guidance aligns with the modality of class names used during the testing phase, ensuring consistency in data distribution, leading to similar mean variances in the cost map

generated during background prediction. Visual semantic guidance outperforms textual semantic guidance on datasets without background classes(see "Visual".). Compared to the latter, unbiased and rich visual semantic guidance provides a superior description, representing the optimal direction for training. Our Dual Semantic Guidance effectively integrates the advantages of both, exhibiting optimal performance on both types of datasets. Secondly, we find that applying MFC loss to textual semantic guidance yields better results than to visual semantic guidance (see "+ MFC Loss"). Compared to visual semantic guidance, language often suffers from feature shift due to the lack of specific descriptions. Therefore, it is necessary to construct a joint optimization framework with both positive and negative samples to constrain the optimization direction. Simultaneously, we observe adverse effects stemming from the modal discrepancies between Guidance Enhance and textual semantic guidance (see "+ Guid. Enhance"). At last, we employ the previous method [6, 9] determines background areas by comparing similarity to a threshold (see "Ours(Thresh 0.4)"). However, its results is less effective compared to ours, proving the effectiveness of predicting background from both spatial and categorical perspectives by fusing feature maps and semantic guidance.

Table 5. Alternative Ways for Region Awareness Attention.

CHG Enc.	Method	w/ BG	w/o BG
✘	CLIP	45.6	39.9
	Gaussian	45.7	40.2
✔	Mask background	1.3	2.7
	class prompt	15.8	13.9
	mask pooling	44.7	39.2
	Ours	**48.3**	**42.8**

4.5. Discussion

Alternative Ways for Region Awareness Attention In Table 5, we evaluated five substitution methods to validate the efficacy of Region Awareness Attention. Firstly,we introduced Gaussian noise(see "Gaussian") to the background regions, which resulted in a decline in performance, because the addition of Gaussian noise does not ensure that the model focuses on the foreground regions. We also attempt to mask the attention of the patch tokens corresponding to the background regions(see "mask background"), and find that this strategy severely impaired the model's feature encoding capabilities. Concurrently, we explore learnable strategy: replacing background tokens with class-specific tokens [53] (see "class prompt"). This strategy is inferior to those of our method, as the lack of training stage prevent these training based methods from effectively adapting. Finally, we refer to the mask pooling strategy of MaskCLIP(see "mask pooling"), whose experimental results are lower than ours, as the low-resolution feature map of CLIP will contain background information and interfere with the instance feature expression.

Table 6. Alternative Ways for Masked Feature Contrastive Loss.

Contrastive	Method	w/ BG	w/o BG
✘	Baseline	45.9	41.2
	MSE loss	38.9	36.7
✔	TCL loss	46.7	41.5
	Ours(Global)	47.7	42.4
	Ours(Avg)	**48.3**	**42.8**

Alternative Ways for Masked Feature Contrastive Loss In Table 6, ablation experiments validate the efficacy of our designed loss function. The ablation study focuses solely on MFC Loss, while other replacement experiments maintain the same BCE Loss. Firstly, we test the MSE Loss (see "MSE loss") which enforces consistency between the overall region embeddings and semantic guidance. The model performed poorly under this loss, as it neglects the information from negative samples. Furthermore, we evaluate the Feature-level TCL loss [6] (see "TCL loss"), which facilitates the efficient computation of features for negative instances. We find that the model's performance is inferior to ours, as this loss function compares whole-image features rather than instance features with the semantic guidance to compute the loss. Additionally, we replace the average pooling with similarity-weighted pooling (see "Ours(Global)"), but this result in a decline. Because this method places greater emphasis on pixel features that are closer to the semantic guidance, thereby underestimating the importance of hard positive pixels for training.

Limitation In the Dual Semantic Guidance Generation, we rely on SAM to generate a multitude of masks for each image, a process that is time-consuming. Furthermore, akin to existing methods in OVSS, our visualization results still suffer from rough boundaries. To boost computational efficiency, we opt for K=5 in K-Means due to its superior performance. Adaptive selection of K is intriguing, which will be considered in future work.

5. Conclusion

Pre-trained Vision-Language Models (VLMs) establish the foundation for recognizing open-vocabulary categories in semantic segmentation tasks. Nonetheless, the dearth of pixel-level annotations in image-text data hinders the provision of thorough and accurate supervisory signals for VLM fine-tuning. This paper introduces Semantic Guidance Generation, which leverages extensive image-text data from visual and textual dimensions, offering comprehensive and precise data to enable VLMs to attain pixel-level segmentation capability. Furthermore, we develop the Dual Semantic Guidance to facilitate efficient model training. Our methodology concurrently and effectively elevates segmentation performance in open-vocabulary scenarios, both with and without the background category.

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